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Over-View Summary:

- **Data Scientist/AI/ML Engineer** with 3+ years of experience in **Large Language Models (LLMs), Machine Learning and Generative AI.**
- Skilled in implementing **ML models** for risk analytics, NLP, and intelligent decision systems, and building scalable **data** pipelines, fine-tuning transformer models. Passionate about leveraging **ML and AI** to solve problems in real-world applications.
- Skilled in building end-to-end ML pipelines with **Python, PyTorch, TensorFlow, Scikit-learn, OpenCV**, and deploying on **Azure and AWS** with CUDA acceleration and **MLOps practices.**
- Proven expertise in building scalable AI systems, optimizing ML workflows, and deploying enterprise-grade **GenAI applications** in multi-cloud environments.
- Extensive expertise in designing, fine-tuning, and deploying **Large Language Models (LLMs)** including GPT, LLaMA, Falcon, and Claude with **Lang Chain, RAG** pipelines, and vector search integration
- Led end-to-end application migrations from AWS and Azure to GCP, modernizing enterprise workloads using **GKE, Cloud SQL, Big Query** and Cloud Functions.
- Hands-on experience in application development with **Python (Fast API), Node.js, and Java** (Spring Boot), deploying scalable micro-services integrated with **ML/AI pipelines.**
- Built predictive ML models for fraud detection, customer churn, demand forecasting, and clinical outcomes, leveraging Random Forest, **XGBoost, and Neural Networks**.
- Strong background in NLP – developed pipelines for document summarization, named entity recognition (NER), sentiment analysis, and medical text de-identification.
- Fine-tuned domain-specific LLMs with LoRA/PEFT techniques, improving accuracy and contextual relevance for enterprise use cases.
- Proficient in Machine Learning (ML) and Deep Learning (DL) with specialization in Natural Language Processing (NLP).
- Proficient in data mining tools like **R, SAS, Python, SQL, Excel, Big Data Hadoop** eco-systems Staff leadership and development
- Strong programming skills in Python, familiar with various python libraries such as **Pandas, NumPy, seaborn, SciPy, MatPlotLib and Scikit-Learn.**
- Experience with **Statistical Analysis, Data Mining** and **Machine Learning** Skills using **R, Python and SQL.**
- Experience in **Agile** delivery environments and all phases **of Software Development Life Cycle** (SDLC).
- Excellent **Communication and presentation skills** along with good experience in communicating and working with various stake holders.

Technical skills:

Programming Languages	Python, R, Java, C#, JavaScript, Dart, SQL.
AI/ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, SHAP, LIME, PDP LLMs & GenAI, ChatGPT-4, Retrieval-Augmented Generation (RAG) pipelines, LangChain, Prompt Engineering, Text Bots, Generative AI workflows.
Data & Visualization	Pandas, NumPy, Matplotlib, Seaborn, Tableau.
Python Libraries	Numpy, SciPy, Pandas, Scikit-learn, Matplotlib, Seaborn, ggplot2, NLP
Algorithms	K-Means, Linear Regression, Non-Negative Matrix Factorization, Decision Tree, Logistic Regression, Naïve Bayes, Random Forest, Matrix Factorization/SVD
Cloud & MLOps	Microsoft Azure (VMs, CUDA), AWS (EC2, S3, RDS, Cognito, ECS Fargate, Secrets Manager), Snowflake, MLflow, Kubeflow, Airflow.
Operating Systems	Windows, Unix, Sun Solaris, Linux

<u>Professional Experience:</u>

Client: M&T Bank, Buffalo, NY. || Duration: JAN 2025-Present
Role: GenAI/ML Engineer
Responsibilities:

- Built and deployed **LLM-powered** applications, including **ChatGPT-4** text bots and Retrieval-Augmented Generation (RAG) pipelines, enabling conversational AI, intelligent automation, and knowledge retrieval across custom datasets.
- Designed and deployed generative AI solutions such as **Retrieval-Augmented Generation** (RAG) pipelines by integrating **GPT-4, LangChain, and Pinecone,** enabling intelligent knowledge retrieval from structured and unstructured enterprise **data** sources, significantly improving search accuracy and response quality for financial domain applications.
- Built an **AI-powered** customer support assistant leveraging **LLMs** that streamlined call centre operations by automating responses to repetitive queries, reducing manual intervention, and enabling real-time escalation for complex cases across multi-channel support platforms using **agentic AI framework LangGraph** with multi agent orchestration.
- Optimized **ML pipelines** using **Python, PyTorch, TensorFlow, and OpenCV**, reducing training/runtime by 30%and debugging overhead by 20%.
- Scaled deep learning workloads on **Azure CUDA VMs**, achieving a 40% increase in training efficiency and enabling multi-dataset experiments.
- Integrated **Hugging Face** Transformers into facial reconstruction pipelines, reducing fine-tuning time by 20%across diverse datasets.
- Built RAG pipelines leveraging **LLMs** for automated asset tagging and code documentation, cutting manual annotation time by 35%.
- Deployed models into production using **CI/CD pipelines, MLflow, and Docker**, ensuring reproducibility and version tracking.
- Engineered scalable, reproducible **Python pipelines for global GHG** datasets and developed cloud-based DSSAT workflows on **GCP with BigQuery** integration and **Vertex AI.**
- Automated end-to-end **ML and GenAI workflows** with **Airflow, Kubeflow, MLflow, Jenkins, and GitHub** Actions, ensuring scalable and reproducible model deployments.
- Implemented robust **data** pipelines using **Apache Airflow and Apache Spark** to automate the ingestion, transformation, and preparation of large-scale financial datasets for model training, ensuring high-quality, reliable, and consistent **data** availability for **GenAI and ML** workloads.
- Developed Python-based **RESTful APIs** to integrate **LLM services** into enterprise applications, ensuring seamless interaction between AI models and internal systems such as CRM, knowledge management, and reporting platforms.
- Configured and managed hybrid cloud networking between **AWS and GCP,** including VPNs, VPC peering, and load balancing, to ensure secure and reliable connectivity across multi-cloud environments supporting **AI/ML** applications.
- Created custom embeddings for financial documents using **Hugging Face Transformers**, enabling semantic search, similarity matching, and document classification to enhance knowledge discovery within large repositories of financial records.
- Automated monitoring and observability workflows by integrating Google Stack driver and Prometheus, setting up real-time alerts and dashboards for model performance, infrastructure health, and latency tracking across distributed systems.
- Supported business stakeholders with ad-hoc ML model performance reporting, preparing clear insights and tailored visualizations to help decision-makers evaluate the effectiveness of deployed AI solutions and align outcomes with organizational goals and delivered in Agile sprints with close coordination across **data**, DevOps, and product teams.

Environment: Python, PyTorch, Hugging Face, Lang Chain, Pinecone, GCP Vertex AI, Big Query, Airflow, Spark, Tableau, Docker, Kubernetes, MLflow, Prometheus, Stack driver.

Company: Cortracker IT Pvt Ltd., INDIA. || Duration: May 2020 – July 2022
Role: AI/ML/Data Scientist
Responsibilities:

- Developed predictive analytics models (Random Forest, XGBoost, Neural Networks) for patient risk stratification, readmission prediction, and disease progression forecasting.
- Designed NLP pipelines to process clinical notes, **EHR data**, and physician documentation, enabling automated named entity recognition (NER) for diagnoses, medications, and procedures.
- Built time-series forecasting models to monitor patient vitals (heart rate, blood pressure, oxygen levels), supporting proactive clinical interventions.
- Implemented data de-identification and PHI masking pipelines using NLP and rule-based methods to ensure HIPAA compliance.
- Applied **Generative AI for clinical** note summarization and automated medical documentation, improving physician productivity.
- Developed drug interaction detection models using graph-based ML techniques to identify adverse medication combinations.
- Created ETL and streaming pipelines with **Apache Kafka, Spark, and Databricks** to process multi-source healthcare data in real time.
- Integrated FHIR standards for healthcare data interoperability, ensuring seamless data exchange between hospital systems and AI pipelines.
- Deployed **ML/GenAI models** on **AWS Sage Maker** enabling scalable inference and retraining pipelines and iteratively refined models in 2-week sprints following Agile Scrum practices..
- Built interactive dashboards **Power BI** to visualize patient risk scores, treatment outcomes, and hospital performance metrics.
- Conducted **A/B testing of AI-powered** patient engagement tools (chatbots, symptom checkers), measuring improvements in communication and appointment adherence.
- Collaborated with clinicians, healthcare providers, and IT teams to translate medical needs into AI/ML solutions that support evidence-based decision-making.

Environment: Python, TensorFlow, PyTorch, Hugging Face Transformers (BERT, BioBERT, GPT models), Scikit-learn, Pandas, NumPy, Apache Spark, Databricks, Kafka, AWS Sage Maker, SQL, Power BI, Docker, Kubernetes, Windows.

Certifications:

- Generative AI with Large Language Models (Coursera – DeepLearning.AI & AWS)
- Machine Learning Specialization by Andrew Ng (Coursera – DeepLearning.AI & Stanford)
- MLOps | Machine Learning Operations (Udemy)
- Lang Chain & LLM App Development: ChatGPT, OpenAI, RAG, Vector DBs (Udemy)